

CENTRAL WISCONSIN, WI, USA

WMO: 726465

Lat: 44.783N

Lon: 89.667W

Elev: 1277

StdP: 14.03

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94890

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-7.3	-20.1	1.9	-9.1	-15.4	2.5	-6.1	24.8	15.6	22.2	17.0	8.7	280	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	86.3	72.3	83.4	70.5	81.2	68.1	74.3	83.0	72.1	80.5	70.1	78.4	9.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	123.3	80.8	69.7	114.5	78.5	66.4	102.1	75.5	38.6	83.8	36.6	79.8	34.8	78.7	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	19.7	17.6	DB	-18.4	91.1	6.4	4.0	-23.0	93.9	-26.7	96.3	-30.3	98.5	-34.9	101.4	(4)
(5)			WB	-17.1	78.1	3.5	2.0	-19.7	79.5	-21.7	80.7	-23.7	81.8	-26.2	83.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	14.8	20.6	29.2	42.7	53.5	63.6	68.5	66.5	59.1	46.4	32.0	20.6	
	DBStd	20.81	11.09	12.22	11.93	9.80	7.83	7.50	5.84	5.79	8.47	8.67	9.66	11.92	
	HDD50	4525	1092	822	653	257	51	2	0	0	19	176	542	911	
	HDD65	8305	1557	1242	1111	669	364	117	31	53	214	580	991	1376	
	CDD50	2058	0	0	7	38	161	410	575	510	291	64	2	0	
	CDD65	360	0	0	0	0	8	74	140	98	37	3	0	0	
	CDH74	3115	0	0	0	26	117	708	1213	709	317	25	0	0	
	CDH80	732	0	0	0	3	17	169	328	143	71	1	0	0	
Wind	WSAvg	8.0	8.5	8.8	9.0	9.3	8.5	7.0	6.6	6.3	7.2	8.3	8.6	8.2	
Precipitation	PrecAvg	32.6	1.1	1.0	1.7	2.9	3.9	4.6	3.7	4.1	3.6	2.8	1.9	1.3	
	PrecMax	38.9	2.7	2.0	4.7	5.0	6.3	8.3	9.5	12.9	8.4	5.6	5.7	3.5	
	PrecMin	25.8	0.3	0.3	0.3	0.9	1.9	1.8	1.2	1.1	0.2	0.4	0.1	0.3	
	PrecStd	3.8	0.6	0.4	1.0	1.1	1.1	1.9	1.9	2.2	1.9	1.3	1.1	0.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	50.3	64.3	78.1	82.1	88.2	90.9	88.1	87.6	77.5	60.6	53.5	
		MCWB	39.4	43.9	54.8	61.4	66.5	72.2	75.7	73.2	71.2	64.7	54.0	51.0	
	2%	DB	36.3	43.3	58.0	71.6	77.3	84.3	87.8	83.6	81.6	71.9	54.5	44.7	
		MCWB	33.0	37.7	49.3	56.0	61.5	70.6	73.7	71.6	67.9	62.2	49.0	40.6	
	5%	DB	32.6	38.9	51.6	64.2	72.8	81.5	83.7	81.1	77.5	65.9	49.8	38.3	
		MCWB	30.9	35.1	46.0	50.6	59.1	68.2	71.2	68.4	65.6	59.0	45.1	34.5	
	10%	DB	29.8	36.3	45.3	59.8	68.7	78.9	81.1	78.6	72.9	61.1	45.7	34.4	
		MCWB	29.1	33.7	40.1	47.7	56.5	66.1	68.5	66.9	62.9	53.8	42.1	32.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	45.7	57.1	63.7	69.0	74.4	78.4	76.4	72.2	68.5	55.4	49.8
			MCDB	43.4	48.8	63.6	76.3	78.9	84.9	89.6	83.6	82.1	75.7	58.2	52.6
2%		WB	33.2	38.3	51.4	57.3	64.4	72.2	75.1	72.8	69.5	63.4	50.0	40.3	
		MCDB	35.7	41.7	55.4	67.0	74.5	82.0	83.9	81.0	80.0	68.9	52.8	44.1	
5%		WB	31.0	36.0	45.9	53.0	61.3	70.0	72.9	70.3	67.2	59.2	45.6	35.2	
		MCDB	32.5	39.0	49.6	61.6	69.8	79.8	80.8	77.9	75.9	65.4	48.8	37.9	
10%		WB	28.7	33.3	39.6	49.0	58.5	67.7	70.6	68.3	64.7	55.0	42.0	32.2	
		MCDB	29.4	36.4	45.2	58.3	66.6	76.1	78.1	75.7	71.2	60.1	46.0	34.6	
Mean Daily Temperature Range	5% DB	MDBR	15.2	16.7	17.6	20.9	21.2	20.8	20.5	19.9	20.8	18.4	14.2	14.2	
		MCDBR	15.9	19.2	23.8	29.2	26.3	23.9	23.0	22.6	24.9	24.9	20.4	18.1	
		MCWBR	13.9	15.3	16.6	17.5	14.8	12.3	12.3	12.4	13.9	16.7	15.5	14.7	
	5% WB	MCDBR	15.5	18.3	21.5	26.6	23.3	22.1	20.7	19.6	21.7	22.9	18.3	16.5	
		MCWBR	13.9	14.9	16.0	17.9	14.3	12.1	12.2	11.9	12.9	16.8	15.2	13.9	
Clear-Sky Solar Irradiance	taub	0.247	0.277	0.316	0.346	0.380	0.381	0.374	0.375	0.349	0.315	0.285	0.255		
	taud	2.402	2.309	2.366	2.377	2.320	2.346	2.377	2.371	2.448	2.531	2.525	2.452		
	Ebn at Noon	283	290	292	291	282	281	282	278	277	275	266	268		
	Edh at Noon	23	30	33	36	40	39	37	36	31	24	21	20		
All-Sky Solar Radiation	RadAvg	528	881	1243	1443	1715	1875	1927	1642	1297	816	511	381		
	RadStd	57	80	132	145	145	134	142	98	108	93	88	54	43	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page